

PRECISION AND RELIABILITY COSTS IN CONTRACTING CONTROL SYSTEMS

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ABSTRACT. We study finite catalogues of open-loop controls that keep assigned initial states within a prescribed Euclidean tolerance of a controlled invariant triangle at every time up to a given horizon. For two planar C^2 maps contracting toward a common origin, matched first derivatives determine a precision spectrum $S(\gamma)$, where the tolerance is $\exp(-\gamma n + o(n))$. We prove that if neither map has a critical point in the triangle, the catalogue rate for failure probability $\exp(-\alpha n + o(n))$ is $\min\{S(\gamma), L_a(\alpha)\}$. Here L_a is the classical binary source-coding reliability function. The probability is uniform physical area, and no coding or feedback strategy is prescribed. The proof transports probability through all feasible transient branches and constructs disjoint positive-area regions that force arbitrary successful approximate inputs to retain specified prefixes. Critical sets of area zero suffice for ordinary precision costs but not for this reliability conclusion: we construct an injective C^2 example with exactly the same full-state catalogue counts as its linear counterpart and a strictly larger reliability cost. This separates qualitative noncollapse from the probability control needed at exponentially small failure budgets. We also identify the first-order matching criterion for common low-precision costs and the cost difference between all states and one fixed nearly full-area compact initial set.

1. INTRODUCTION

Invariance entropy quantifies the number of control signals needed to maintain a state constraint; see Kawan [8]. When every input contracts toward a common origin, a fixed positive environment tolerance can lead to zero asymptotic information cost. Two requirements reveal more of the control geometry: let the tolerance decrease with the horizon, and let the allowed initial-state failure probability decrease as well. We ask what dynamical conditions preserve the resulting precision–reliability cost.

Our catalogues assign an entire input to an actual initial state. For $u \in \{0, 1\}^{\mathbb{N}}$, the successful states satisfy $\text{dist}(F_u^k x, Q) < \delta$ at every $0 \leq k \leq n$. An input may leave and reenter Q , but cannot omit any intermediate constraint. Initial states have uniform area distribution on the same triangle $Q = \{0 \leq s \leq 1, |z| \leq s\}$ throughout the paper. The success set may depend on the horizon. This differs from covering a single fixed compact set of almost full area.

The main positive result, Theorem 2.1, concerns nonlinear maps whose first derivatives have angular branch proportions $a_0 + a_1 = 1$ and radial factors a_i^β , $\beta > 1$. The maps need not be globally injective, triangular, or radially autonomous. They make Q controlled invariant and contract toward the origin in a common norm. If both derivatives are nonsingular on Q , then for every finite precision exponent $\gamma \geq 0$, every failure exponent $\alpha > 0$, and all sequences with these exponents, the minimum successful catalogue has rate $\min\{S(\gamma), L_a(\alpha)\}$. The function S is the full-state precision spectrum, while $L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p)$ is the classical binary reliability function, with I_a the relative entropy. The formula is not a new source-coding function. The assertion is its validity for unrestricted approximate controls and physical area under a nonlinear transient.

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Theorem 2.2 shows why a stronger probability hypothesis is needed than for ordinary precision costs. We construct an injective C^2 pair whose critical sets in Q have area zero, with no positive-area collapse. Its full-state catalogue count agrees with the matched linear count for every horizon and tolerance. Nevertheless, at fixed precision and failure exponents its reliability rate has lower limit strictly larger than the displayed formula. A single transient sends relatively large physical regions into small angular regions requiring many different controls. Those physical regions have vanishing total area, but cannot be discarded at the specified exponential failure budget.

The connection to the true minimum catalogue requires two estimates. First, all finite feasible transient branches have a common inverse-area bound when there are no critical points on Q . Together with local complete-word area bounds, this dominates physical word-family probabilities by reference product weights. It provides a reliable catalogue using exactly feasible prefixes, without charging an $O(\delta_n)$ escape layer to the smaller failure budget ϵ_n . Second, bounded-run words are realised on a whole band of positive radii and thickened to disjoint regions of area proportional to their angular product. Every point has a physical wrong-control margin. Thus any successful approximate input must retain the word while its radial scale exceeds the tolerance. These two statements, proved in Section 7, connect classical type counting to actual probability and arbitrary inputs.

The precision results underlying this comparison also explain initial-set dependence. Critical sets of area zero allow area transport on one fixed regular compact window. Under matching, all positive-area compact initial sets have rate γ/β up to βh , where $h = -\sum_i a_i \log a_i$. One fixed compact set of arbitrarily nearly full area has rate $\min\{h, \gamma/\beta\}$ at every finite exponent, whereas the full-state rate grows to $\log 2$. Within the prescribed first-order family, unequal radial powers already separate these costs for every sufficiently small positive exponent. These results describe ordinary precision costs; they do not replace the probability estimates required for reliability.

1.1. Related work. The finite-horizon full-cover object is outer invariance complexity [1, Section 2.2]. This framework already permits noninjective continuous maps. Stabilization entropies [3, 4] concern related costs with prescribed decay or practical stabilization requirements. Our constraints are environment distances at all times, with both tolerance and allowed initial-state failure varying with the horizon.

Wang, Huang and Sun [12] develop measure-theoretic invariance entropies using invariant-partition cylinder probabilities. Their conjugacy theorem transports the initial measure together with the system. Colonius [10] connects quasi-stationary metric invariance entropy to coder-controllers and proves entropy relations under measure-preserving transformations. Wang and Huang [11] study spanning-based measure-theoretic entropy and inverse variational principles. We do not claim that probability-dependent control entropy or measure transport is new. Here one must control the mass actually covered by any approximate input, at shrinking tolerance and exponentially small failure, under uniform physical area. The finite-transient probability domination and the positive-area forced regions supply this control.

Nie, Wang and Huang [9] derive measure entropy by convex duality from invariance pressure. Chen, Huang and Zhong [2] use fixed invariant-partition cylinders and time-power weights for entropy dimensions. These tools describe an already specified pressure or cylinder family; their use for the present catalogue requires a quantitative relation to the actual covered sets. We prove that relation rather than identifying a feasible coding with an optimal approximate catalogue.

Local graph estimates, summable distortion, product stopping scales and typical-set arguments are classical tools [7, 6, 5]. Cost-threshold pressure and Bowen equations for invariant partitions are developed in [13]. Our type estimates and reliability optimisation are elementary instances of these established ideas. The independent control steps are the

complete-region estimates, the all-branch probability transport, and the physical forcing of arbitrary successful inputs. All required estimates are proved below, without assuming a symbolic partition or a pressure formula for the true catalogue.

Section 2 states the probability result, its obstruction, and the precision results used for comparison. Sections 3–6 prove the local geometry and precision bounds. Section 7 proves the main probability theorem; Section 8 constructs the obstruction. The final examples and scope distinguish the hypotheses of these statements.

2. THE CATALOGUE AND MAIN RESULTS

2.1. Catalogue and precision. Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone.

Let $\mu = \text{area}|_Q$, which is a probability since $\text{area}(Q) = 1$. For $0 < e < 1$, define

$$(2.2) \quad r_e(n, \delta; Q) = \min \left\{ \#\mathcal{U} : \mu \left(\bigcup_{u \in \mathcal{U}} A_u(n, \delta; Q) \right) \geq 1 - e \right\}.$$

The minimum is over finite catalogues of unrestricted binary inputs. It exists because the full-cover count is finite. A failure sequence has $0 < e_n < 1$, $e_n \rightarrow 0$ and $-\log e_n/n \rightarrow \alpha > 0$. Both sequences may be nonmonotone. The successful physical subset in (2.2) may change with n ; it is not a fixed compact initial set.

2.2. First-order data and global assumptions. Fix

$$(2.3) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.4) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

The standing assumptions on the actual maps are:

- (1) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 maps, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (2) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (3) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.5) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

- (4) Each critical set

$$(2.6) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . The fourth supplies noncollapse along feasible finite transients; it allows folds and requires no condition on critical points outside Q . Radial motion may depend on angle. No accumulated distortion, word-area bound or catalogue comparison is assumed.

Both matched and mismatched subfamilies are nonempty. For the linear maps A_i , the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det A_i = r_i^2/a_i > 0$.

For words write

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. Define

$$(2.7) \quad \chi_R = - \sum_i a_i \log r_i > h, \quad \gamma_0 = - \log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

2.3. Probability costs. Write H_b for binary entropy. In the matched case $\beta_0 = \beta_1 = \beta$, the angular cost and spectrum are

$$(2.8) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(2.9) \quad S(\gamma) = \max_{p \in [0,1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

$$(2.10) \quad I_a(p) = \chi(p) - H_b(p), \quad L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p).$$

The function I_a is relative entropy with respect to (a_0, a_1) ; L_a is the classical binary source-coding reliability function.

Theorem 2.1 (Precision–reliability transfer). *Assume the standing hypotheses with $\beta_0 = \beta_1 = \beta$, and strengthen the critical-set condition to*

$$(2.11) \quad \det DF_i(x) \neq 0 \quad (x \in Q, i = 0, 1).$$

For every fixed such system, every finite $\gamma \geq 0$, every $\alpha > 0$, and every precision and failure sequence of those exponents,

$$(2.12) \quad \lim_{n \rightarrow \infty} \frac{\log r_{e_n}(n, \delta_n; Q)}{n} = \min\{S(\gamma), L_a(\alpha)\}.$$

No global injectivity or global inverse is required.

For a finite catalogue, enumerate its inputs and assign a successful state to its first successful input. The sets are Borel, so this is a measurable initial-state encoder. A fixed-length index sent once at time zero has minimum bit length $\lceil \log_2 r_e \rceil$; the asymptotic bit rate is (2.12) divided by $\log 2$. There are no later messages. This interpretation does not identify the catalogue rate with a causal feedback data-rate or a channel capacity. The bit conversion is a definition-level consequence.

Theorem 2.2 (A null-critical transient changes reliability). *For $a_0 = 2^{-10}$, $a_1 = 1 - 2^{-10}$ and $\beta = 4096$, there is a fixed injective C^2 pair satisfying the standing matched hypotheses, with critical sets of area zero, such that*

$$(2.13) \quad r_F(n, \delta; Q) = r_A(n, \delta; Q) \quad (n \geq 1, \delta > 0).$$

It maps no positive-area subset of Q into an area-zero set. Nevertheless, for $\gamma = -\beta \log(a_0 a_1)$ and $\alpha = 1/8$, every pair of sequences with these exponents satisfies

$$(2.14) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \frac{1}{2} \log 2 > \min\{S(\gamma), L_a(\alpha)\}.$$

Condition (2.11) is a sufficient condition for (2.12). The counterexample rules out replacing it merely by critical sets of area zero in a universal transfer theorem. It does not show that every system with critical points fails, or give a necessary and sufficient probability criterion.

2.4. Ordinary precision costs and fixed initial sets. The following results use only the standing null-critical hypothesis. They both describe the baseline precision cost and distinguish it from the exponentially reliable cost above.

Theorem 2.3 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.15) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.16) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.17) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.4 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.3, put $a = a_* \in (0, 1/2)$, $c_b = -\log(1 - a)$ and $d_b = \log((1 - a)/a) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.18) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.19) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.20) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.*

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

Theorem 2.5 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$ and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem 2.6 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(2.21) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

2.5. The physical coverage estimate. The following estimate bounds the mass covered by one arbitrary input, without selecting an exact coding. It supplies the lower rate for positive-area initial sets in both cases. Its regular compact set also provides the single near-full set used in the upper bounds. These are fixed-loss estimates: their $C_T \delta$ term is not assigned to an exponentially small probability budget.

Theorem 2.7 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ and every $0 < \varepsilon < \text{area}(K)$, one can choose a compact $T \subset K \setminus \{0\}$ with $\text{area}(T) > \text{area}(K) - \varepsilon$ and $C_T < \infty$ so that for every $e \in (0, h)$ there is $C_{T,e} < \infty$ such that*

$$(2.22) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T \delta + C_{T,e} \left[e^{-(h-e)m} + \delta e^{(\chi_R - h + 2e)m} \right]$$

for every input u , $n \geq J$, $0 \leq m \leq n - J$ and $0 < \delta \leq 1$. The integer depends only on the fixed system. The set and C_T precede e, n, δ, u ; the second constant may also depend on e . In particular, for $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

3. LOCAL FEASIBLE GEOMETRY

We first control complete feasible regions near the origin. The angular and radial products describe different scales, so they remain separate throughout this section. The first-exit estimate will then apply to any input, even one that is only approximately feasible.

Use (s, y) with $z = sy$, only for $s > 0$. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\bar{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*$, and shrink $\rho > 0$ so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i . In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the generous common constant B ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example $|N_i| \leq 3$ on this rectangle; quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the displayed choice of B dominates all the constants in (3.3). These are derived local inequalities, not extra hypotheses.

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (3.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (3.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (3.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$;

the second sums the angular derivatives. They are independent product estimates. Only in the matched case is $P_w = \lambda_w^\beta$. Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 10.1 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

3.2. The first true exit. For a fixed input, partition its covered states by the first time at which its trajectory actually leaves Q . Up to that time the prefix is exactly feasible, so the preceding interval estimates apply. The permitted tolerance restricts the states that can survive this first exit.

Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least cs_0P_w , by (3.3),(3.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most $C\delta/(s_0P_w)$. By (3.6), its initial-angle preimage has length at most

$$C\delta\lambda_w/(s_0P_w).$$

After integration against $s_0 ds_0$, its physical area is at most

$$(3.9) \quad C\delta\lambda_w/P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

4. FINITE TRANSIENTS AND ACTUAL COVERED AREA

Common contraction gives a uniform time of entry into the local region. To transport its area estimates back to the initial set, we remove a small area once and retain all feasible transient choices on the remaining compact window. This is where the null-critical-set hypothesis is used; a global inverse is not required.

For a finite word w , let

$$(4.1) \quad D_w = \{x \in Q : F_{w|k}x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write w^- for the word with its last letter removed. At a first true exit under w , all derivative inputs still lie in Q , so the relevant domain is D_{w^-} . The last output need not belong to Q .

Lemma 4.1 (Null pullbacks along feasible prefixes). *For every Lebesgue null set $B \subset \mathbb{R}^2$, every control i , and every finite word w ,*

$$(4.2) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

For each nonempty word the compact set

$$(4.3) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

also has area zero.

Proof. The inverse function theorem supplies an open inverse chart at every point of $Q \setminus \mathcal{C}_i$. A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its C^1 inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of B is null. The omitted $\mathcal{C}_i$ is null by (2.6). A Borel null superset handles an arbitrary Lebesgue null set.

For $w = vi$, exact feasibility gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (4.2), including the empty word. Finally, on D_{w-} the chain rule expresses $\det DF_w$ as a product of determinants evaluated at states in Q . If the factor at time j vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1}\mathcal{C}_{w_j}.$$

This is null by (4.2). Their finite union contains Z_w , which is compact by continuity and closedness of the feasibility domain. $\square$

Choose $J \geq 1$, depending only on the fixed system, so that

$$(4.4) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible J -prefix ends in $Q_\rho = Q \cap \{s \leq \rho\}$.

Lemma 4.2 (One regular compact window). *For every positive-area compact $K \subset Q$ and $0 < \varepsilon < \text{area}(K)$, there is a compact $T \subset K \setminus \{0\}$, with $\text{area}(T) > \text{area}(K) - \varepsilon$, and $L_T < \infty$ such that*

$$(4.5) \quad \text{area}(T \cap D_{w-} \cap F_w^{-1}B) \leq L_T \text{area}(B)$$

for all $1 \leq |w| \leq J$ and all Lebesgue measurable $B \subset \mathbb{R}^2$. On the same set, for every $e \in (0, h)$ there is $D_e \geq 1$ such that every local exactly feasible tail v , after any exactly feasible J -prefix from T , satisfies

$$(4.6) \quad \begin{aligned} D_e^{-1}e^{-(h+e)|v|} &\leq \lambda_v \leq D_e e^{-(h-e)|v|}, \\ D_e^{-1}e^{-(\chi_R+e)|v|} &\leq P_v \leq D_e e^{-(\chi_R-e)|v|}. \end{aligned}$$

The set and L_T precede e , every input, tolerance and horizon.

Proof. For $d > 0$, the total weight $\sum \lambda_v$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. Indeed, for a Bernoulli variable X of mean a_0 , the second derivative of $\log \mathbb{E}e^{t(X-a_0)}$ is a tilted variance, at most $1/4$. Its value and derivative at zero vanish, so it is at most $t^2/8$. Exponential Markov with $t = 4d$ and $t = -4d$ gives the bound. By (3.7), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli for rational $d > 0$ gives a Borel null set $B_{\text{loc}} \subset Q_\rho$, including the origin, outside which all feasible length- m prefixes have zero frequency tending uniformly to a_0 . Overlaps do not invalidate this union bound. Controlled invariance supplies at least one feasible prefix at every length.

Lemma 4.1 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1}B_{\text{loc}})$$

is null. The finite union $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$ is a compact null set. On $Q \setminus B_J$, the maximum frequency deviation over all feasible transient choices and all their feasible length- m tails tends to zero. It is measurable: for each m there are finitely many words and their complete feasibility domains are closed. Egorov on K , followed by removal of $B_J \cup Z_J \cup \{0\}$ and compact inner approximation, gives T of the stated area with uniform convergence.

Allocate less than $\varepsilon/2$ loss to Egorov and less than $\varepsilon/2$ to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (4.6); finitely many early lengths are absorbed in D_e . This selects one set before all precision choices, rather than a set for each exponent.

For a word w with $1 \leq |w| \leq J$, the compact set $M_w = T \cap D_{w-}$ is disjoint from Z_w . The composition F_w is therefore regular at every point of M_w . Cover it by finitely many open inverse charts $U_{w,k}$, with compact closures inside regular inverse charts. Change of variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1}B) \leq \left(\sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty M_w 's, to obtain (4.5). Chart overlaps only enlarge the bound; every local inverse branch meeting M_w is included. $\square$

The lemma provides an inverse-area bound on a fixed regular window, not a Jacobian lower bound on all Q or a global multiplicity bound. Its constants may depend on the prescribed area loss. The upper catalogues below still cover every actual state, including the critical states omitted from this window.

Proof of Theorem 2.7. Choose J, T, L_T from Lemma 4.2. Let $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$. For $0 < \delta \leq 1$, the tubes around the three edges of Q give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

If a state in $T \cap A_u(n, \delta; K)$ first truly exits at $j \leq J$, put $w = u|j$. Its earlier states lie exactly in Q , and it belongs to

$$T \cap D_{w-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (4.5), its class has area at most $L_T C_Q \delta$. Summing over the J possible first-exit times gives

$$(4.7) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in Q ; no nonsingularity condition is needed at the last output, which may be outside.

For the remaining states, $\sigma = u|J$ is exactly feasible. Transport them by F_σ to Q_ρ , and use (4.5) to pull back endpoint area estimates. If the next m prescribed digits are exactly feasible for at least one transported state, (3.7) and (4.6) bound the area of the survival class by $C_{T,e} e^{-(h-e)m}$. If no such state exists, that class is empty. For a first true local exit at time $j \leq m$, the prescribed $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (3.9), (4.5) and both products in (4.6), is at most

$$C_{T,e} \delta e^{(\chi_R - h + 2e)(j-1)}.$$

Since $\chi_R > h$, the sum is bounded by a fixed multiple of $\delta e^{(\chi_R - h + 2e)m}$. Together with (4.7) and exact survival, this proves (2.22). A switch between feasible codes was never counted as an exit. Every covered state satisfies the constraint at its first true exit, regardless of subsequent reentry. $\square$

5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

The area estimate concerns positive-area mass. For the full constraint we also need actual states whose controls cannot be replaced by approximate alternatives. We construct them at one fixed positive radius for each run bound, then force prefixes while their physical margin remains larger than the tolerance.

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that, for every $0 < s \leq s_L$, every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at (s, sy_0) . At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The initial-angle centre can be chosen Lipschitz in s , with a constant uniform over the sequences. The radius bound and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. The inverse reference maps are $f_0(y) = a_0y - a_1$, $f_1(y) = a_1y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L + 1$ digits. Since $f_0(y) + 1 = a_0(y + 1)$, $1 - f_1(y) = a_1(1 - y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular recursion has yet been imposed. The scalar map $s \mapsto s R_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its equation is exactly $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, not a formal symbolic substitution. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_* d_L/(4a^*)$. Both wrong angular excesses are then at least $d'_L = a_* d_L/(2a^*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. The same estimates hold for every initial radius $0 < s \leq s_L$. For a fixed candidate sequence, scalar contraction gives $\sup_k |s_k(s, y) - s_k(s', y)| \leq |s - s'|$. If $\kappa_L < 1$ is the common contraction constant of (5.3), the fixed-point equations therefore give

$$(5.5) \quad \|y^v(s) - y^v(s')\|_\infty \leq \frac{a^* B}{1 - \kappa_L} |s - s'|.$$

This proves the uniform physical realisation, continuity, and genuine-infeasibility assertions. $\square$

The preceding margin is angular. Its physical size decreases with the radius, so a long exact word need not remain distinguishable at an assigned tolerance. The next lemma chooses a shorter prefix and compares its radial scale directly with that tolerance.

Lemma 5.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(5.6) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (3.6), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$. The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(5.7) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (3.8), regardless of later reentry. Thus the approximate input must agree with all m_n witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (5.6). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

6. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first construct catalogues by following feasible inputs until the radius is small enough for a common tail. This construction applies under all the standing assumptions. We then prove the matched spectrum using the physical prefix lower bound and the covered-area estimate. The reference tree provides an upper bound; its leaves need not all be realised by actual states.

Lemma 6.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_{Rp}\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary

states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta / M \leq \delta / (4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (2.5) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (6.1). $\square$

6.1. Upper bound for the matched spectrum. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. Then $P_w = \lambda_w^\beta$, $\chi_R = \beta h$, and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(6.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. Endpoint types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No nonempty fixed-radius slice or full terminal image is assumed.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Shorter physical prefixes give the lower bound. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 5.2 is now $c_L = \beta\chi_L$. Fix L first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (6.4) give the zero limit. This proves (2.15).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.3, apply Theorem 2.7 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $e \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Each term on the right of (2.22) is then at most a constant times $e^{-(h-e)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - e) \min\{1, \gamma/(\chi_R + e)\}.$$

Since $\chi_R = \beta h$ here, letting $e \downarrow 0$ gives the lower bound in (2.16); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.15). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N + 1) \exp \left(N \max_{|p - a_0| \leq d_N} H_b(p) \right) = e^{(h + o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (6.5) also gives upper rate γ/β . The lower bound already proved gives (2.17). The construction of T used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.3 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.4. Reparameterise the maximum by the frequency q of the smaller branch $\alpha = a_*$. This only changes the optimisation variable, not the dynamics or control assignments. Then $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1 - q) \log((1 - q)/(1 - \alpha)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = \alpha$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at α .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > \alpha$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $\alpha < 1/2$, $c_b + d_b/2 > c_b + d_b\alpha = h$. The three pieces match at the thresholds, proving (2.18)–(2.19) and the low-side assertion of Theorem 2.3. Subtracting (2.17) gives (2.20). Strict increase of H_b on $(\alpha, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

Corollary 6.2 (Matched linear systems). *If $\beta_0 = \beta_1 = \beta$, the pair (A_0, A_1) has all the conclusions of Theorems 2.3 and 2.7.*

Proof. The angular maps are $(y+a_1)/a_0$ and $(y-a_0)/a_1$. Their feasible domains $[-1, a_0-a_1]$ and $[a_0-a_1, 1]$ cover the fibre. Put $t = \max_i t_i < 1$, choose $\Lambda = 1 + 4t/(1-t)$ and set $v = t(1 + 1/\Lambda) < 1$. Then $V(A_i x) \leq \max\{\max_i r_i, v\}V(x)$ globally. The remaining hypotheses are immediate. $\square$

7. PHYSICAL PROBABILITY AND EXPONENTIAL RELIABILITY

Throughout this section the powers match, $r_i = a_i^\beta$, and (2.11) holds. We first prove probability domination on the whole triangle. The lower bound then uses regions with known positive physical mass, rather than only the individual witnesses needed for the full-state precision spectrum.

7.1. All feasible transient branches. Let J and ρ be as in Section 4, so every exactly feasible J -prefix ends in Q_ρ . Recall D_w , the complete exactly feasible domain, and w^- , the word with its last digit removed.

Lemma 7.1 (Whole-triangle probability domination). *There is a fixed $K_J < \infty$ such that for every $1 \leq |w| \leq J$ and measurable $B_0 \subset \mathbb{R}^2$,*

$$(7.1) \quad \mu(D_{w^-} \cap F_w^{-1} B_0) \leq K_J \text{area}(B_0).$$

Consequently, for every family $\mathcal{V}$ of words of a common length N ,

$$(7.2) \quad \mu\left(\bigcup_{\substack{|\sigma|=J \\ v \in \mathcal{V}}} D_{\sigma v}\right) \leq C_0 \sum_{v \in \mathcal{V}} \lambda_v, \quad C_0 = \max\{1, 2^J K_J C_A \rho^2\}.$$

The constants are independent of N , the word family, tolerance, failure budget, and horizon.

Proof. Compactness gives $d_0 = \min_{i,x \in Q} |\det DF_i(x)| > 0$. On D_{w^-} all inputs to the derivative factors of F_w belong to Q , even when its last output leaves Q . Thus $|\det DF_w| \geq d_0^{|w|}$ there. Cover this compact domain by finitely many local inverse charts for F_w , with compact closures inside regular charts. Change of variables bounds the pulled-back area in each chart by its inverse-determinant supremum times $\text{area}(B_0)$. Sum over the charts and maximise over the finitely many words of length at most J . All inverse branches meeting the domain are included; their overlap only enlarges the upper bound. No global inverse is used.

An exactly feasible σ sends D_σ into Q_ρ . Feasibility of the tail v places the endpoint in its actual complete region $\mathcal{I}_v$, whose area is at most $C_A \rho^2 \lambda_v$ by (3.7). A possible endpoint at zero has a null pullback by (7.1). Pull back each region and sum over all at most 2^J transients and all v . This proves (7.2). $\square$

This is domination, not equality of word probabilities or a claim that feasible regions form a partition. Applying (7.1) to the first true exit during the transient would bound its area by $JK_J C_Q \delta$. We do not discard this area at failure cost e_n : δ_n can be exponentially larger than e_n . The next construction uses exactly feasible prefixes.

7.2. Two catalogue upper bounds. Fix $\varepsilon > 0$, $n > J$, and $N = n - J$. Keep every exactly feasible length- n prefix σv , with $|\sigma| = J$, whose tail zero frequency p satisfies $I_a(p) \leq \alpha + \varepsilon$. Empty complete regions may be omitted. Append any infinite input to each kept horizon prefix. Every state not covered has an exactly feasible length- n prefix by controlled invariance, and all such prefixes from that state have discarded tails. By Lemma 7.1, its total probability is at most

$$(7.3) \quad C_0 \sum_{I_a(j/N) > \alpha + \varepsilon} \binom{N}{j} e^{-N\chi(j/N)} \leq C_0(N+1)e^{-N(\alpha+\varepsilon)} < e_n$$

for large n . Here $\binom{N}{j} \leq e^{NH_b(j/N)}$ follows from the j -th term of the binomial probability sum. The catalogue has at most

$$(7.4) \quad 2^J(N+1)e^{NL_a(\alpha+\varepsilon)}$$

members. The fixed transient and polynomial factors have zero rate. Continuity of L_a , proved below, gives upper rate $L_a(\alpha)$.

Independently, the full-state stopping construction in Lemma 6.1 gives upper rate $S(\gamma)$. It assigns an actual feasible prefix to each state; after a threshold leaf its common tail satisfies every remaining constraint. We need not combine the two catalogues: choose the cheaper one. Therefore

$$(7.5) \quad \limsup_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \leq \min\{S(\gamma), L_a(\alpha)\}.$$

7.3. A fixed positive mass gives the low-side lower bound. Take one regular compact $T \subset Q$ of area greater than $3/4$ from Theorem 2.7. For fixed $0 < \varepsilon < h$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta h + \varepsilon} \right\rfloor \right\}.$$

Then $\delta_n \leq e^{-(\beta h + \varepsilon)m_n}$ and, using $\chi_R = \beta h$, (2.22) bounds the area covered inside T by one input by a fixed multiple of $e^{-(h-\varepsilon)m_n}$. A successful catalogue covers more than $1/2$ area inside T eventually. The union bound and $\varepsilon \downarrow 0$ give

$$(7.6) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \min\{h, \gamma/\beta\}.$$

Since $S(\gamma) = \gamma/\beta$ for $\gamma \leq \beta h$ and $L_a(\alpha) \geq h$, this matches (7.5) on the entire low side. At $\gamma = 0$, the upper bound and $r_{e_n} \geq 1$ give the zero limit. Fixed loss is used here to retain a positive successful mass, not to construct an upper bound with exponentially small failure.

7.4. Positive-area regions forcing complete prefixes.

Lemma 7.2 (Thickened physical words). *For every integer $L \geq 2$ there are $\tau_L, E_L > 0$ such that any family $\mathcal{W}$ of distinct words of a common length m , each admitting an infinite extension with runs at most L , has pairwise disjoint measurable regions $E_w \subset Q$ with*

$$(7.7) \quad \mu(E_w) = \tau_L \lambda_w.$$

Every point of E_w is exactly feasible under w . If $n \geq m$ and $2\delta < E_L P_w$, every input whose actual successful set $A_u(n, \delta; Q)$ meets E_w has prefix w . All constants are fixed before m, n, δ and the word family.

Proof. For each word choose a bounded-run extension $v(w)$. Lemma 5.1 gives a common radius bound $s_L > 0$, continuous centres $y_w(s)$ for $0 < s \leq s_L$, and wrong-control excess $d'_L > 0$. Its construction gives centre angles at distance at least $\Delta_L = 3d_L/4$ from both boundaries at every step, where $d_L = 2a_*^{L+1}$. Put $K_0 = 1/a_* + 3B\rho$ and choose

$$0 < \epsilon_L \leq \min\{1/(2C_A), \Delta_L/(4C_A), d'_L/(4K_0C_A)\}.$$

Define

$$(7.8) \quad E_w = \{(s, sy) : s_L/2 \leq s \leq s_L, |y - y_w(s)| \leq \epsilon_L \lambda_w\}.$$

Fix the initial radius. On the feasible component between the centre and any feasible angle, (3.6) bounds the change in the angle after each prefix $w|j$ by

$$(7.9) \quad |Y_{w|j}(y) - Y_{w|j}(y_w(s))| \leq C_A \epsilon_L \lambda_w / \lambda_{w|j} \leq D_L, \quad D_L = C_A \epsilon_L.$$

If the complete interval $I_w(s)$ ended within the proposed initial interval, continuity would make an initial or intermediate angular constraint attain an endpoint. Its centre is at distance Δ_L , while $D_L \leq \Delta_L/4$, a contradiction. The radii stay positive and below $\rho < 1$, so no radial constraint terminates the interval. Thus the entire interval is feasible. This argument applies even when the complete terminal image is truncated; it does not assume that image is onto.

At each prefix the thickened state and its centre lie on one image graph with logarithmic radial slope at most one. Hence their log radius ratio has absolute value at most $D_L \leq 1/2$, and their radius difference is at most $2\rho D_L$. The alternative angular map changes by at most

$$(1/a_* + B\rho)D_L + B(2\rho D_L) = K_0 D_L \leq d'_L/4.$$

Every thickened point therefore retains a wrong-control excess at least $d'_L/2$ at every prefix, as well as selected-control feasibility. Continuity of the centres makes E_w measurable. Its actual area is

$$2\epsilon_L \lambda_w \int_{s_L/2}^{s_L} s \, ds = \frac{3\epsilon_L s_L^2}{4} \lambda_w; \quad \tau_L = 3\epsilon_L s_L^2/4.$$

Two distinct equal-length words cannot share a point in their regions: at their first difference one selected control would be wrong, contradicting feasibility under both words.

If an arbitrary input first differs at $j \leq m$, the preceding trajectory is exactly the selected one. Its radius is at least $C_R^{-1}(s_L/2)P_{w|j-1} \geq C_R^{-1}(s_L/2)P_w$. The wrong output has radial factor at least $r_*/2$ and angular excess at least $d'_L/2$. Thus

$$(7.10) \quad |z_j| - s_j \geq E_L P_w, \quad E_L = \frac{s_L r_* d'_L}{8C_R} > 0.$$

Either $z - s$ or $-z - s$ has this positive excess. Each function has Euclidean Lipschitz constant $\sqrt{2}$ and is nonpositive on Q , so the distance is at least $E_L P_w / \sqrt{2} > \delta$. The constraint is violated at that time. Later reentry cannot remove the violation. This proves the all-point forcing assertion. $\square$

7.5. Rare physical types and completion of the proof. Suppose $c = \gamma/\beta > h$. Fix an interior frequency p with

$$(7.11) \quad \chi(p) < c, \quad I_a(p) < \alpha.$$

Choose $j_L/L \rightarrow p$ and use the length- L blocks starting with zero, ending with one, and containing j_L zeros. Write

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L), \quad I_L = \chi_L - B_L.$$

The binomial bounds proved in Section 6 give $B_L \rightarrow H_b(p)$, $\chi_L \rightarrow \chi(p)$ and $I_L \rightarrow I_a(p)$. Fix L large enough that $\beta\chi_L < \gamma$ and $I_L < \alpha$. Put $k = \lfloor n/L \rfloor$ and $m = Lk$. The M_L^k concatenations, extended by a fixed allowed block, have runs at most L . Lemma 7.2 gives disjoint regions of equal mass and equal radial product,

$$\mu(E_w) = \tau_L e^{-\chi_L m}, \quad P_w = e^{-\beta\chi_L m}.$$

The physical margin exceeds $2\delta_n$ eventually because its logarithmic ratio has rate $\gamma - \beta\chi_L > 0$. One arbitrary input can then cover points in at most one region. A catalogue of R inputs successful on $1 - e_n$ area must obey

$$(7.12) \quad R \geq M_L^k - \frac{e_n}{\tau_L e^{-\chi_L m}} = M_L^k \left(1 - \frac{e_n}{\tau_L e^{-I_L m}} \right).$$

The parenthesis tends to one, since $\alpha > I_L$. Take the horizon limit with L fixed, then let $L \rightarrow \infty$. This proves

$$(7.13) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \sup_{\chi(p) < c, I_a(p) < \alpha} H_b(p) \quad (c > h).$$

To evaluate this lower bound, let $a = a_* < 1/2$ and use q for the frequency of the smaller branch. Put $c_b = -\log(1 - a)$, $d = \log((1 - a)/a)$, and $J_a = I_a(1/2) > 0$ in this relabelled notation. Then $\chi(q) = c_b + dq$, $h = \chi(a) = H_b(a)$, and Corollary 2.4 gives the three pieces of S . Strict convexity of relative entropy, together with monotonicity of H_b on $[0, 1/2]$, gives

$$(7.14) \quad L_a(\alpha) = \begin{cases} H_b(q_\alpha), & 0 < \alpha < J_a, \quad I_a(q_\alpha) = \alpha, \quad a < q_\alpha < 1/2, \\ \log 2, & \alpha \geq J_a. \end{cases}$$

This also proves continuity, including at J_a . For $c > h$ set $q_c = \min\{1/2, (c - c_b)/d\}$, and set $q_\alpha = 1/2$ if $\alpha \geq J_a$. Every $a < q < \min\{q_c, q_\alpha\}$ satisfies (7.11). Approaching its endpoint in (7.13) yields

$$\liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq H_b(\min\{q_c, q_\alpha\}) = \min\{S(\gamma), L_a(\alpha)\}.$$

Together with (7.5) and (7.6), this proves Theorem 2.1. The common-cost endpoint $c = h$ uses the fixed-mass argument, while the saturation and reliability endpoints use strict types approaching from below. Integer rounding changes m by less than fixed L ; the transient changes N by fixed J . The strict gaps in (7.3) and (7.12) absorb every $o(n)$ fluctuation in both sequences. No constants are required to be uniform as $L \rightarrow \infty$, or over different fixed systems.

Corollary 7.3 (Linear and fixed-failure specialisations). *The matched linear pair has rate (2.12) for every stated pair of sequences. Under the weaker standing null-critical hypotheses, each fixed $e \in (0, 1)$ instead satisfies*

$$(7.15) \quad \lim_n n^{-1} \log r_e(n, \delta_n; Q) = \min\{h, \gamma/\beta\}.$$

Proof. The linear pair is nonsingular on Q , so Theorem 2.1 applies. For fixed failure, choose the single compact set in Theorem 2.3 with area greater than $1 - e$; its catalogue gives the upper rate in (7.15). For the lower rate choose a regular window T of area greater than $(1 + e)/2$. A catalogue with failure at most e covers at least $\text{area}(T) - e > 0$ area there. The fixed-mass proof above, which needs only Theorem 2.7, gives the lower rate. $\square$

8. A TRANSIENT PRESERVING FULL COSTS BUT CHANGING RELIABILITY

We prove Theorem 2.2. The construction alters only the first step from Q . It remains injective and has no positive-area collapse. Its small target regions have physical preimages whose probabilities decay too slowly to omit at the required failure budget.

8.1. A flat increasing Cantor interpolation. Fix the parameters of the theorem and put $\lambda = a_0 a_1$. The inverse reference maps are $f_0(y) = a_0 y - a_1$ and $f_1(y) = a_1 y + a_0$. For the control blocks 01 and 10, write

$$B_0 = f_0 \circ f_1 = \lambda y + a_0^2 - a_1, \quad B_1 = f_1 \circ f_0 = \lambda y + a_0 - a_1^2.$$

Their invariant interval has endpoints and length

$$(8.1) \quad A = \frac{a_0^2 - a_1}{1 - \lambda}, \quad B = \frac{a_0 - a_1^2}{1 - \lambda}, \quad \ell = B - A = \frac{2\lambda}{1 - \lambda}.$$

Indeed $A + 1 = 2a_0^2/(1 - \lambda) > 0$ and $1 - B = 2a_1^2/(1 - \lambda) > 0$, so $[A, B] \subset (-1, 1)$. The maps take this interval to its left and right portions of relative length $\lambda < 1/2$. The corresponding Cantor set $\mathcal{C}$ consists of angles with infinite block controls 01, 10.

On $J_0 = [-1/2, 1/2]$ use the source maps

$$b = 2/5, \quad h_0(y) = by - (1 - b)/2, \quad h_1(y) = by + (1 - b)/2.$$

Their Cantor set $\mathcal{D}$ is null, because the total level- k interval length is $(2b)^k \rightarrow 0$. For $\sigma \in \{0, 1\}^k$ put

$$J_\sigma = h_{\sigma_1} \circ \cdots \circ h_{\sigma_k}(J_0), \quad Z_\sigma = B_{\sigma_1} \circ \cdots \circ B_{\sigma_k}([A, B]).$$

Within each family these closed intervals are separated by positive gaps. Their lengths are b^k and $\ell\lambda^k$, respectively. Both address maps preserve order and are unique, including endpoints. The target Cantor set is null as $(2\lambda)^k \rightarrow 0$.

We now construct a strictly increasing C^2 homeomorphism $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$(8.2) \quad g(y) = y \ (|y| \geq 1), \quad g([-1, 1]) = [-1, 1], \quad g(J_\sigma) = Z_\sigma, \quad g'(y) = 0 \iff y \in \mathcal{D}.$$

Let $E(t) = e^{-1/t}$ for $t > 0$ and zero otherwise, and set

$$\phi(t) = E(t)E(1 - t), \quad \Sigma(t) = \frac{\int_0^t \phi(v) dv}{\int_0^1 \phi(v) dv} \quad (0 \leq t \leq 1),$$

extended by zero and one outside that interval. The function is smooth, strictly increasing on $(0, 1)$, and flat at both endpoints. Map each point of $\mathcal{D}$ to the point of $\mathcal{C}$ with the same address. On a source gap (u, v) whose corresponding target gap is (U, V) , define

$$(8.3) \quad g(y) = U + (V - U)\Sigma((y - u)/(v - u)).$$

The ordered endpoint correspondence and increasing interpolation give the full interval identity $g(J_\sigma) = Z_\sigma$. At level k , the gap lengths are $(1 - 2b)b^k$ and $(1 - 2\lambda)\ell\lambda^k$. For $j = 1, 2$ the gap derivatives obey

$$(8.4) \quad |g^{(j)}| \leq C_j(\lambda/b^j)^k.$$

Since $\lambda < 2^{-10} < b^2$, both bounds tend to zero with level. Define q_1, q_2 by these derivatives inside gaps and by zero on $\mathcal{D}$. They are continuous, by (8.4) for high-level gaps and endpoint flatness for the remaining finite collection.

These candidate derivatives are actual derivatives. The integral of q_1 over a gap is its target gap length. Nullity of $\mathcal{C}$ and order preservation therefore imply $g(x) = A + \int_{-1/2}^x q_1(y) dy$ on J_0 ; inside a gap include the partial integral of (8.3). Moreover

$$\sum_{\text{gaps}} \int |q_2| \leq C \sum_{k \geq 0} (2\lambda/b)^k < \infty.$$

Each complete-gap integral of q_2 is zero, and a partial-gap integral equals $q_1(x)$. Since $\mathcal{D}$ is null, $q_1(x) = \int_{-1/2}^x q_2(y) dy$. The fundamental theorem of calculus gives $g' = q_1$ and $g'' = q_2$. Thus g is C^2 on J_0 , flat to second order on $\mathcal{D}$, with positive first derivative on all gaps.

For completeness, the exterior joins can be made without altering these properties. On $[L, R]$ with prescribed increment $T > 0$, set $d = \min\{(R - L)/4, T/4\}$ and $v(x) =$

$E(x - L)E(R - x)$. For the left join use $c(x) = 1 - \Sigma((x - L)/d)$, and for the right use $c(x) = \Sigma((x - R + d)/d)$. In either case $\int c \leq d < T$. Integrate the positive interior density

$$(8.5) \quad q(x) = c(x) + \frac{T - \int_L^R c}{\int_L^R v} v(x).$$

For $(-1, -1/2, A + 1)$ start at $g(-1) = -1$; for $(1/2, 1, 1 - B)$ start at $g(1/2) = B$. The identity endpoint has derivatives 1, 0, and the Cantor endpoint has derivatives 0, 0. Extending by identity beyond $[-1, 1]$ therefore gives a global C^2 map satisfying (8.2). It is strictly increasing and onto, and $|g(y) - y| \leq 2$ globally. The map is fixed independently of every horizon and both sequences.

8.2. The actual maps and all dynamical assumptions. Set

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

Define

$$H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + s\psi(s)[g(z/s) - z/s]), & s > 1/4, \end{cases}$$

and set $F_i = A_i \circ H$. The smooth function ψ is zero on $s \leq 1/4$ and one on $s \geq 1/2$. The quotient is used only away from zero; its needed derivatives are locally bounded at every joining point, and flatness of ψ gives global C^2 maps. Near the origin H is identity, so the required first derivatives agree exactly.

On every fibre the angular map $g_s(y) = (1 - \psi(s))y + \psi(s)g(y)$ is an increasing homeomorphism of $[-1, 1]$ onto itself. Therefore $H(Q) = Q$, and one of the two linear controls is feasible at $H(x)$ for each $x \in Q$. Its radial factor is less than one. This proves controlled invariance including every boundary and the origin.

Put $m_* = \max_i \{a_i^\beta, a_i^{\beta-1}\} < 1$, $\Lambda = 1 + 10m_*/(1 - m_*)$, and $V = \max\{\Lambda|s|, |z|\}$. The bound $|H^z - z| \leq 2|s|$ gives, on the whole plane,

$$V(Hx) \leq (1 + 2/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently the common contraction factor can be taken as

$$\theta = m_*(1 + 1/\Lambda)(1 + 2/\Lambda) \leq m_*[1 + 5/\Lambda] < (1 + m_*)/2 < 1.$$

This applies to arbitrary input tails outside Q as well.

For $s > 1/4$ the determinant is

$$\det DH = 1 - \psi(s) + \psi(s)g'(z/s), \quad \det DF_i = a_i^{2\beta-1} \det DH;$$

for $s \leq 1/4$ the first determinant equals one. Thus the critical set in Q is exactly $\{(s, sy) : 1/2 \leq s \leq 1, y \in \mathcal{D}\}$. It has area zero by Fubini in physical coordinates. Local inverse charts off that null set show that every null target set has a null pullback in Q . No positive-area set is collapsed to a null set. Each fibre map is strictly increasing and identity outside $[-1, 1]$, so H is globally bijective. It is proper, since its first coordinate is unchanged and $|H^z - z| \leq 2|s|$. Thus it and both F_i are global homeomorphisms. They are not asserted to be diffeomorphisms; the displayed critical points are genuine.

The fixed radial factors satisfy

$$a_1^\beta = (1 - 2^{-10})^{4096} < e^{-4} < 1/4, \quad a_0^\beta < 1/4.$$

After the first input from Q , the radius lies below $1/4$, regardless of whether its angle is feasible. Every subsequent input therefore uses the identity part of H , even along an approximate trajectory that has left Q . For every input and every $j \geq 1$,

$$(8.6) \quad F_u^j x = A_u^j Hx \quad (x \in Q).$$

At time zero both x and Hx belong to Q . It follows that $A_u^F(n, \delta; Q) = H^{-1}A_u^A(n, \delta; Q)$ on Q . Since $H(Q) = Q$, full covers correspond, proving (2.13). Probability covers correspond only after transporting μ to $H_*\mu$, which is absolutely continuous but is not uniform area. This is consistent with measure-transport invariance results; it is not a counterexample to them.

8.3. Positive-area forced families and the strict rate gap. Every infinite sequence of blocks 01, 10 has runs at most two. The reference endpoint-distance identities used in (5.1) give $|y_j| \leq 1 - d$ at all shifted positions, with $d = 2a_*^3$. For a chosen zero, the current angle is at most $a_0 - a_1 - a_0d$, so the alternative one gives at most $-1 - a_0d/a_1$; the symmetric bound holds for a chosen one. Thus every wrong-control angular excess is at least $d_* = (a_*/a^*)d > 0$. This holds on each complete interval Z_σ through its $2k$ -digit block prefix: its endpoints have such bounded-run extensions, and the linear prefix maps are monotone. No symbolic Cantor subset replaces the whole interval in this assertion.

For each $\sigma \in \{0, 1\}^k$ define an actual physical region

$$E_{\sigma,k} = \{(s, sy) : 1/2 \leq s \leq 1, y \in J_\sigma\}.$$

Here $\psi = 1$ and $g(J_\sigma) = Z_\sigma$. The regions are disjoint and each has mass $(3/8)b^k$, so their union has mass $(3/8)(2b)^k$. Every point requires its selected $2k$ -digit control prefix at sufficiently small tolerance. To check this for an arbitrary input, let its first differing digit be $j \leq 2k$. Equation (8.6) and $s \geq 1/2$ give the preceding physical radius at least $\frac{1}{2}\lambda^{\beta k}$: a partial block product before $2k$ is no smaller than the complete product λ^k . The wrong output radial factor is at least a_*^β and its angular excess is at least d_* . Therefore

$$(8.7) \quad |z_j| - s_j \geq c_0 \lambda^{\beta k}, \quad c_0 = \frac{1}{2} a_*^\beta d_* > 0.$$

Its distance from Q is at least this value divided by $\sqrt{2}$. For $k = \lfloor n/2 \rfloor$ and $\gamma = -\beta \log \lambda$, the logarithmic ratio to δ_n has positive rate $-\frac{1}{2}\beta \log \lambda$. Every successful input thus has the full selected prefix, for every point in the region. Later reentry cannot remove the constraint at this first wrong step.

One catalogue input can consequently cover points of at most one region. A catalogue of R inputs with failure at most e_n obeys

$$(8.8) \quad R \geq 2^k \left(1 - \frac{8e_n}{3(2b)^k} \right).$$

Since $-\frac{1}{2} \log(2b) = \frac{1}{2} \log(5/4) < 1/8 = \alpha$ by $\log(1+x) < x$, the parenthesis tends to one. Thus the lower limit is at least $\frac{1}{2} \log 2$. Both strict exponent gaps absorb all the allowed $o(n)$ fluctuations.

Finally $S(\gamma) = \log 2$: at $p = 1/2$ the threshold ratio $\gamma/[\beta\chi(p)]$ is two. For $q = 1/16$,

$$H_b(q) = \frac{1}{4} \log 2 + \frac{15}{16} \log(16/15) < \frac{1}{4} \log 2 + \frac{1}{16} < \frac{1}{2} \log 2,$$

and

$$I_a(q) > \frac{5}{8} \log 2 - H_b(q) > \frac{3}{8} \log 2 - \frac{1}{16} > \frac{1}{8},$$

using $\log 2 > 1/2$. Relative entropy is strictly increasing for $q > a_0$; its sublevel $I_a(q) \leq 1/8$ therefore has right endpoint below $1/16$. Binary entropy is increasing up to $1/2$, so $L_a(1/8) < H_b(1/16) < \frac{1}{2} \log 2$. This proves the strict inequality (2.14) and completes Theorem 2.2.

The example leaves the full-state finite-horizon counts unchanged but changes the probability weights of forced physical regions. Its own accurate reliability spectrum and the existence of its reliability limit are not determined here; the strict lower-limit gap suffices for the obstruction. Absolute continuity and null-set pullback preservation alone do not control exponential tail masses.

9. MISMATCH AND THE COMMON-COST CRITERION

We now return to the standing assumptions with $\beta_0 \neq \beta_1$. The same physical estimates apply, but the angular and radial products no longer have a common-power identity.

Proof of Theorem 2.6. First consider the upper rate on Q . For every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 6.1 cross $P_w \leq \delta_n/M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(9.1) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This proves the upper rate γD through an actual full-time catalogue, without asserting that the reference tree is necessary.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(9.2) \quad H_b(p_*) = D c_R(p_*).$$

Choose $j_L/L \rightarrow p_*$ and use the blocks in Lemma 5.2, with $c_L = c_R(j_L/L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 6 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma/c_L < 1$ therefore gives lower rate $t B_L$. Take its supremum first, then let $L \rightarrow \infty$. Equation (9.2) gives the lower rate γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (2.21).

For any fixed positive-area compact K , take the set T in Theorem 2.7, and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Every term of (2.22) is then bounded by a constant times $e^{-(h-e)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - e)/(\chi_R + e)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $e \downarrow 0$. This proves the asserted universal lower bound $\gamma h/\chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 6.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta/M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_e}{\chi_R - e},$$

by (4.6). At this length, $\lambda_w \geq D_e^{-1} e^{-(h+e)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_e e^{(h+e)\ell}$ such words. Sum over these lengths and the at most 2^J transients, and include one input for the origin. For fixed $e \in (0, h)$ this gives

$$(9.3) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h + e}{\chi_R - e} [q_{\delta_n} + c_{\max} + \log D_e] + O_e(\log(1 + q_{\delta_n})) + O_e(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $e \downarrow 0$ gives upper rate $\gamma h/\chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before e , the precision exponent, the sequence, and the horizon.

Finally let $b_i = r_i^D$. The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h/\chi_R$, completing the theorem. $\square$

Proof of Theorem 2.5. If $\beta_0 = \beta_1 = \beta$, Theorem 2.3 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem 2.6 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The criterion concerns precisely the first derivatives (2.4), nonsingularity almost everywhere in Q , controlled feasibility, and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

10. APPLICATIONS OF THE CRITERION

The following nonsingular example has nonlinear, truncated feasible images. It satisfies the reliability hypotheses and cannot be reduced to the matched linear pair by the stated constraint-preserving simultaneous conjugacies.

10.1. Inward boundary images and empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1-t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the single explicit family

$$(10.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (2.4). For the stated V ,

$$(10.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(10.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside $(-1, 1)$, and their overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. This proves controlled invariance and the actual overlap geometry. Theorem 2.3 applies.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(10.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns $I_0^m(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_0^m$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in Q .

There is also a constraint-preserving conjugacy obstruction. For (10.1), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(10.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of Q into ∂Q and whose one control sends the whole upper edge into ∂Q , both corresponding intersections contain a continuum. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (10.5) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (10.2) gives a global factor at most $67/90 < 1$. Corollary 2.4 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

11. CONSEQUENCES AND SCOPE

The precision and reliability costs use the same actual initial-state space and the same all-time distance constraints, but require different probability control. Under matched first derivatives and null critical sets, all positive-area compact sets share the low-precision rate, while one fixed nearly full-area set omits the high-cost rare states. Its selection precedes every finite precision exponent and sequence. The full-triangle spectrum continues to grow above βh and saturates at the distinct threshold $-(\beta/2) \log(a_0 a_1)$. Unequal first-order radial powers separate the costs already near zero within the prescribed matrix family.

At exponentially decreasing failure, these rare states cannot be treated as a fixed area loss. With no critical points on Q , the finite-transient transport bound holds on the whole probability space. Together with positive-area forced prefixes, it gives the exact precision–reliability rate. In the Cantor transient, the ordinary full-state catalogues are identical to the linear ones, but physical tail probabilities change their exponent. Thus preservation of an ordinary precision spectrum does not imply preservation of reliability. The nonsingular condition is sufficient, and the weaker null-critical condition is insufficient for a universal probability transfer; no necessary and sufficient classification is claimed.

The results concern binary planar maps with the specified first-order matrices, a controlled invariant triangle, and common norm contraction toward the origin. Radial feedback, vanishing local overlap, and truncated feasible images are permitted. The norm condition is not pairwise distance contraction. The reliability theorem fixes uniform physical area on Q ; transporting a measure is a different assertion. No causal communication capacity, optimality of a prescribed strategy, high-precision mismatch spectrum, or classification of every compact initial set is deduced. The conjugacy obstruction is limited to the stated inward example and Q -preserving simultaneous control maps. The counterexample's exact reliability rate remains outside the claims.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

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